



Technical Service Bulletin

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Cylinder Head Changes

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This service/parts topic provides a history of the changes made to the QSK45/60 cylinder head. Also provided is updated parts information, which will be required during in-shop rebuilds of QSK45/60 cylinder heads.

The Tribaloy intake valve, Part Number 4016438, is faced with a durable, wear-resistant material: Tribaloy. This valve is used on all QSK45/60 applications, including haul-truck applications, and on applications with valve beat-in problems. When the intake valve is used with rotators, it reduces the rate of intake valve beat-in. These high-load, constant-speed applications include prime-power and base-load generator sets, marine applications, some drilling machines, and hydraulic excavators.

⚠ CAUTION ⚠

Do not use the Tribaloy intake valve, Part Number 4016438, without the Tribaloy intake valve insert, Part Number 4080326, and valve rotator, Part Number 3085961. Do not use the Tribaloy intake valve in the exhaust valve position. Incorrect use of this valve can cause engine damage.

The valve head thickness of the Tribaloy intake valve, Part Number 4016438, **must** be measured with the valve head thickness tool, Part Number 3823800. The intake valve head thickness tool, Part Number 3375940, **must not** be used on the Tribaloy intake valve. If it is used on the valve, the Tribaloy facing could be ground too thin.

The material of the intake valve, Part Number 4016438, is Tribaloy 400, which is denoted with a "K."

Intake valve stem seal, Part Number 3170134, changed to Part Number 3408282. This prevents excessive carbon build-up.

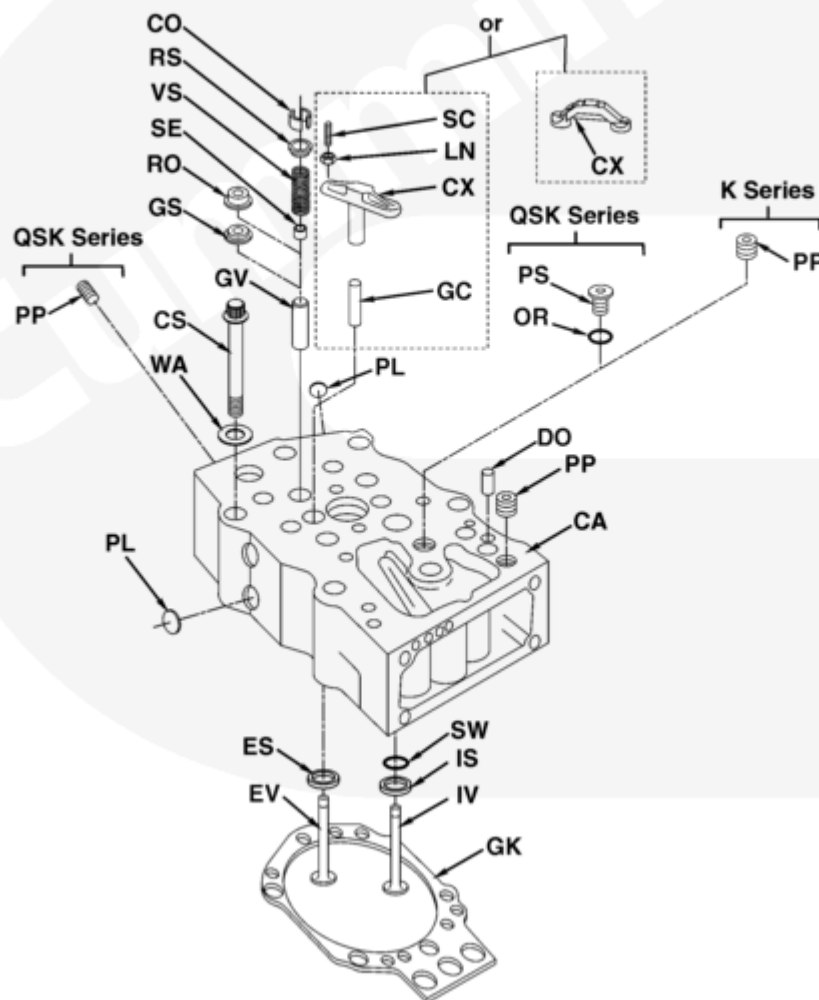
Pyromet with Stellite Face Exhaust Valve

Because of exhaust valve failures, early valves, Part Number 3090469, were changed to Pyromet, Part Number 4065481, with a Stellite face on all QSKV engines. This valve has greater high-temperature fatigue strength with resistance to face pitting and valve beat-in and will provide improved durability. The engine serial number first is 33146911, July 2000.

Valve Identification

There are two methods to identify intake and exhaust valves; **always** use the part number for identification when possible. Part numbers are stamped on the valve stem just above the collet groove. A letter code, cast onto a recess on the combustion face of the valve head, often can be used when the part number is illegible.

K19, K38, K50, KTA38GC, QSK19, QSK23, QSK45, QSK60, QSK78, and QSK19, QSK38 and QSK50 Electronically Actuated Injector



02-0011hp

The material of the exhaust valve, Part Number 4065481, is Pyromet with a Stellite facing, which is denoted with an "L." Before engine serial number 33146911, valve identification was with an "S," which stood for Stellite 6 facing.

Valve Leakage Testing

Vacuum leakage-testing cup, Part Number 3823852, and seal, Part Number 3823853, were released primarily for K Series engine cylinder heads. The vacuum leakage-testing cup and seal will **not** determine if there is a leak between the valve insert and cylinder head.

Valve Spring Identification

The present valve spring used on the QSK45/60 cylinder head is a chromium silicon valve spring, Part Number 3085962. This valve spring is identified by a blue-painted mark and is classified as a beehive style of spring. Following are additional valve spring reuse specifications: A bent, broken, or nicked valve spring **must not** be reused.

If a valve spring with a notch worn in the first active coil is reused, the ends of the valve spring will wear into the first coil, creating a worn notch in the material. If a worn notch can be seen or felt, the valve spring **must not** be reused. Often a valve spring that has these will make a clicking noise. To check for the noise, compress the valve spring by hand until the end of the first coil is completely collapsed. A large notch will cause a clicking noise. Listen for the noise at both ends of the valve spring. Refer to Table 1 below for valve spring load reuse guidelines.

Table 1, Spring Load/Height Verification Values
Load at 71.882 mm [2.83 in] = 77.111 kg ±4.083 kg [170 lb ±9 lb]
Load at 57.226 mm [2.253 in] = 158.757 kg ±8.165 kg [350 lb ±18 lb] (-8% relaxation) = 138.346-kg [305-lb] minimum load

Valve Spring Rotator

The valve spring rotator is designed to decrease valve guide wear. It lets the valve spring rotate the valve naturally as the valve opens and prevents the valve from rotating back into the original position. The valve spring rotator fits between the cylinder head and the valve spring and adds approximately 10.3 mm [0.406 in] to the valve spring assembly height. When installing a QSKV valve spring rotator, Part Number 3085961, a rotation check of the valve spring rotator in the free state can **not** be done.

When turning the valve spring rotator by hand, it can be difficult to turn or can **not** turn at all. Even if a valve spring rotator is difficult to turn by hand, it is **not** "seized"; such difficulty is common. Slight dimensional variations of components can make some valve spring rotators seem tight; but when the valve spring load is applied, the valve spring rotator compresses, which provides more clearance between the collar and the body of the valve spring rotator, allowing it to rotate freely.

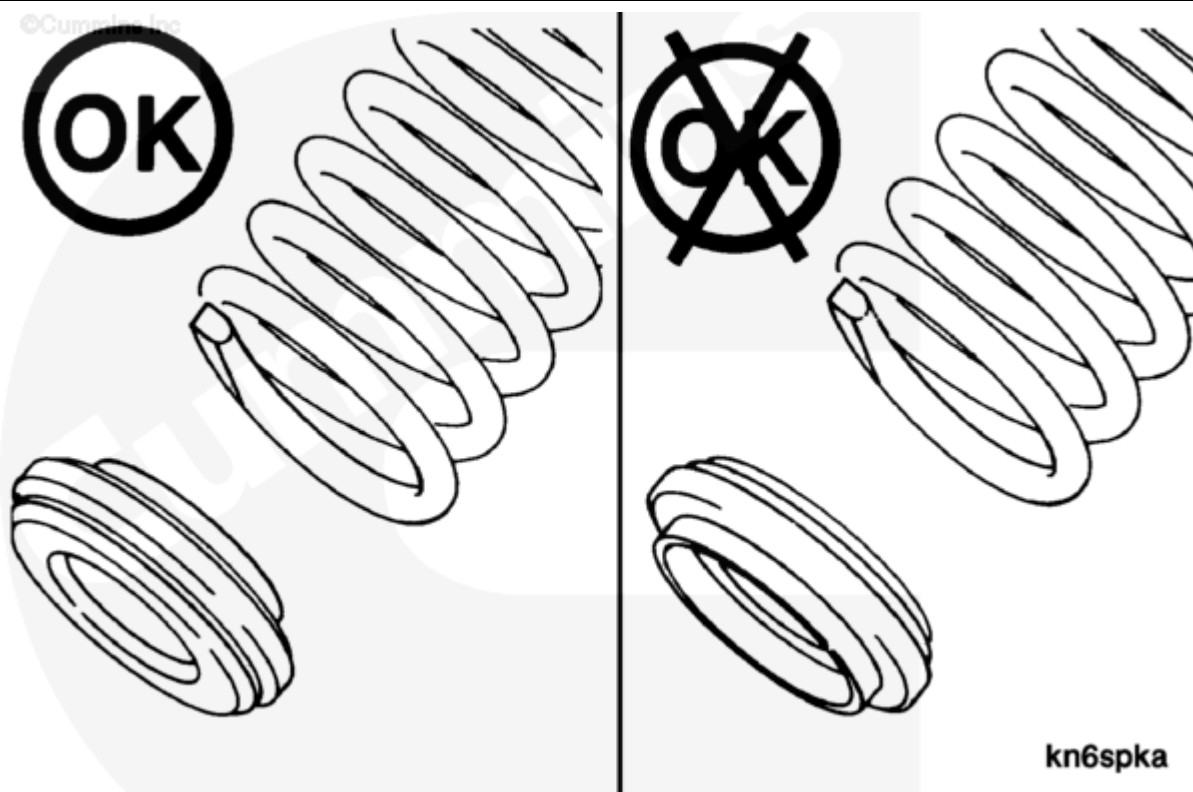


Fig. 1, Rotator Installed over the Valve Guide with the Spring-Pilot Flange Face-up

Valve Crossheads and Guides

Cummins has released a stemless valve crosshead, Part Number 3085964, for use on all QSKV 45/60 engines. No adjustment is required, and it can be installed with the small pocket on either valve. Adjustment of the rocker lever to crosshead lash remains unchanged.

Intake valve seat, Part Number 4080223, changed to Part Number 4080326 to correct valve-to-insert overhang issues. Exhaust valve seat, Part Numbers 3171702 to 4080333, introduced to correct valve-to-insert overhang issues, is engine serial number first 33149408. The intake seat is made of Tribaloy 400, and the exhaust seat is made of Stellite 3.

Table 2, QSK45 and QSK60 Cylinder Head Part Numbers

Item	Present Part Number	Old Part Number
Head assembly	4080358	4080224
Head casting	4080225	N/A
Dowel pin	3900257	N/A
Exhaust seat	4080333	3171702
Exhaust valve	4065481 (4025070*)	N/A
Exhaust valve stem seal	4080343	N/A
Expansion plug	3007635	N/A
Expansion plug (1/4-in)	203933	N/A
Expansion plug (7/8-in)	205401	N/A
Intake seat	4080326	4080223

Intake valve	4016438 (3800955*)	N/A
Intake valve stem seal	3408282	3170134
Valve collet	3081534	N/A
Valve guide	3170135	N/A
Valve spring	3085962	N/A
Valve spring retainer	3331752	N/A
Valve spring rotator	3085961	N/A
Pipe plug	3278039	N/A
*With collets N/A = Not applicable		

Document History

Date	Details
2019-3-25	Non-Product Problem Solving (PPS)